Forest Fire Prediction Using Random Forest and Satellite Image Analysis(CNN)

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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1. Abstract:

Forest fires have become one of the most serious environmental challenges, causing widespread damage to forests, wildlife, ecosystems, and human lives. Detecting fires at an early stage and predicting their occurrence can help reduce these losses and improve disaster management. This project presents a forest fire prediction system that combines satellite image analysis with machine learning and deep learning techniques. The proposed approach uses a **Convolutional Neural Network (CNN)** to extract meaningful features from satellite images, enabling the automatic identification of fire-related patterns such as smoke, flames, and burned vegetation. These extracted features are then combined with environmental factors, including temperature, humidity, rainfall, wind speed, and vegetation indices, to improve prediction accuracy.

A **Random Forest** classifier is used as the prediction model because of its ability to handle complex datasets, reduce overfitting, and provide reliable classification results. The model is trained using historical satellite imagery and weather data, allowing it to distinguish between fire-prone and non-fire-prone regions. Before training, the dataset undergoes preprocessing steps such as data cleaning, normalization, image resizing, and feature extraction to improve model performance. The overall system is evaluated using standard performance metrics, including accuracy, precision, recall, F1-score, and confusion matrix.

The integration of CNN-based image analysis with the Random Forest algorithm provides a robust framework for early forest fire prediction. Compared with traditional methods that rely only on weather conditions or manual observation, this approach offers faster detection and improved predictive capability. The proposed system can support forest departments, environmental agencies, and disaster management authorities by providing timely warnings and assisting in preventive decision-making. By leveraging advances in artificial intelligence and satellite remote sensing, this work contributes to the development of an efficient, cost-effective, and scalable solution for reducing the environmental and economic impact of forest fires

2. INTRODUCTION:

Forest fires are among the most destructive natural disasters, causing severe damage to forests, wildlife, biodiversity, and human settlements. Every year, millions of hectares of forest land are destroyed due to natural causes such as lightning and prolonged drought, as well as human activities like unattended campfires, agricultural burning, and discarded cigarettes. The increasing frequency of forest fires, driven by climate change and rising temperatures, highlights the need for reliable and efficient fire prediction systems.

Traditional forest fire monitoring methods mainly depend on manual surveillance, watchtowers, and ground-based sensors. Although these methods are useful, they are often time-consuming, expensive, and limited in their ability to monitor large forest areas. Recent developments in artificial intelligence (AI), machine learning (ML), deep learning (DL), and satellite remote sensing have created new opportunities for detecting and predicting forest fires more accurately and at an earlier stage.

3. Software Requirements:

Operating System: Windows 10/11 or Ubuntu Programming Language: Python 3.x
IDE: Visual Studio Code / PyCharm / Jupyter Notebook
Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas Database: SQLite (Optional)
Dataset: NASA MODIS / Sentinel-2 Satellite Images Version Control: Git
Hardware: Intel Core i5, 8 GB RAM, 50 GB Storage (GPU optional)

Signature of the Guide Course Instructor